Discussion Section 2
STAT 132 , Winter 2025

$\pi(\theta) = \text{Beta}(2, 200)$

1. Suppose that the proportion θ of defective items in a large shipment is unknown and that the prior distribution of θ is the beta distribution with parameters α 2 and β 200. If 100 items are selected at random from the shipment and if three of these items are found to be defective, what is the posterior distribution of θ? $\sim \text{Beta}(2+3, 200+97)$

2. Suppose that a random sample of 100 observations is to be taken from a normal distribution for which the value of the mean θ is unknown and the standard deviation is 2, and the prior distribution of θ is a normal distribution. Show that no matter how large the standard deviation of the prior distribution is, the standard deviation of the posterior distribution will be less than 1/5. ① $X \sim N(\theta, 2^2)$ $\theta \sim N(\mu, \sigma^2)$ $\theta|X \sim N(\mu^*, \sigma^{*2})$

likelihood

3. Suppose that the time in minutes required to serve a customer at a certain facility has an exponential distribution for which the value of the parameter θ is unknown and that the prior distribution of θ is a gamma distribution for which the mean is 0.2 and the standard deviation is 1. If the average time required to serve a random sample of 20 customers is observed to be 3.8 minutes, what is the posterior distribution of θ?

4. Suppose that a random sample of size n is taken from the Bernoulli distribution with parameter θ, which is unknown, and that the prior distribution of θ is a beta distribution for which the mean is μ_0. Show that the mean of the posterior distribution of θ will be a weighted average having the form $\gamma_n \bar{X}_n + (1 - \gamma_n)\mu_0$, and show that $\gamma_n \to 1$ as $n \to \infty$.

5. Suppose that the number of defects in a 1200-foot roll of magnetic recording tape has a Poisson distribution for which the value of the mean θ is unknown, and the prior distribution of θ is the gamma distribution with parameters $\alpha = 3$ and $\beta = 1$. When five rolls of this tape are selected at random and inspected, the numbers of defects found on the rolls are 2, 2, 6, 0, and 3. Using squared error loss, what is the Bayes estimate of θ?

$X = \frac{\theta^x}{x!} e^{-\theta}$

$\pi(\theta) = \text{Gamma}(3, 1)$

$\pi(\theta|X) = \text{Gamma}(3+13, 1+5)$

Posterior Mean

mean

Absolute error loss

→ Median

Z-table

$P(X < 0) = 0.5$

likelihood

$$P(\theta|Y) = \frac{P(Y|\theta)\cdot P(\theta)}{P(Y)} \quad \begin{matrix} \to \text{prior} \\ \to c \end{matrix}$$

$P(\theta|Y)$: posterior; θ: para.; Y: data

2. $\sigma^{*2} = \left(\frac{1}{\sigma^2} + \frac{100}{2^2}\right)^{-1} = \left(\underbrace{\frac{1}{\sigma^2} + 25}_{>25}\right)^{-1} < \frac{1}{25} = \left(\frac{1}{5}\right)^2$

3. $t|\theta \sim \exp(\theta) = \theta e^{-\theta t}$

$\pi(\theta) = \text{gamma}(25, 125)$

$t \to \begin{cases} n = 20 \\ \bar{t} = 3.8 \end{cases}$ $\quad \sum t = n\cdot\bar{t}$

$\pi(\theta|t) = \text{gamma}(25+20, 125+20\times 3.8)$

4. $X \sim \text{Ber}(\theta)$ $\quad \pi(\theta) = \text{Beta}(a, b)$ $\quad \frac{a}{a+b} = \mu_0$

$\pi(\theta|X) = \text{Beta}(a^*, b^*)$

$$\frac{a^*}{a^*+b^*} = \frac{a+\sum X_i}{a+\sum X_i + b + n - \sum X_i} = \frac{a+\sum X_i}{a+b+n}$$

$$= \underbrace{\frac{a}{a+b+n}}_{\frac{a}{a+b}\cdot\frac{a+b}{a+b+n}} + \underbrace{\frac{\sum X_i}{a+b+n}}_{\frac{n\bar{X}_n}{a+b+n} = \underbrace{\left(\frac{n}{a+b+n}\right)}_{\gamma_n}\cdot\bar{X}_n}$$

$$\frac{a}{a+b}\cdot\frac{a+b}{a+b+n} = \underbrace{\left(\frac{a+b}{a+b+n}\right)}_{1-\gamma_n}\cdot\mu_0$$